

Introduction

Around **1%** of the world's water is stored in both freshwater and liquid form (USBR, 2020). Out of this 1%, **1.3 trillion gallons** of water are used for manufacturing each year in the textile field (Drew & Yehounme, 2017), constituting **one fifth of total wastewater pollution**.

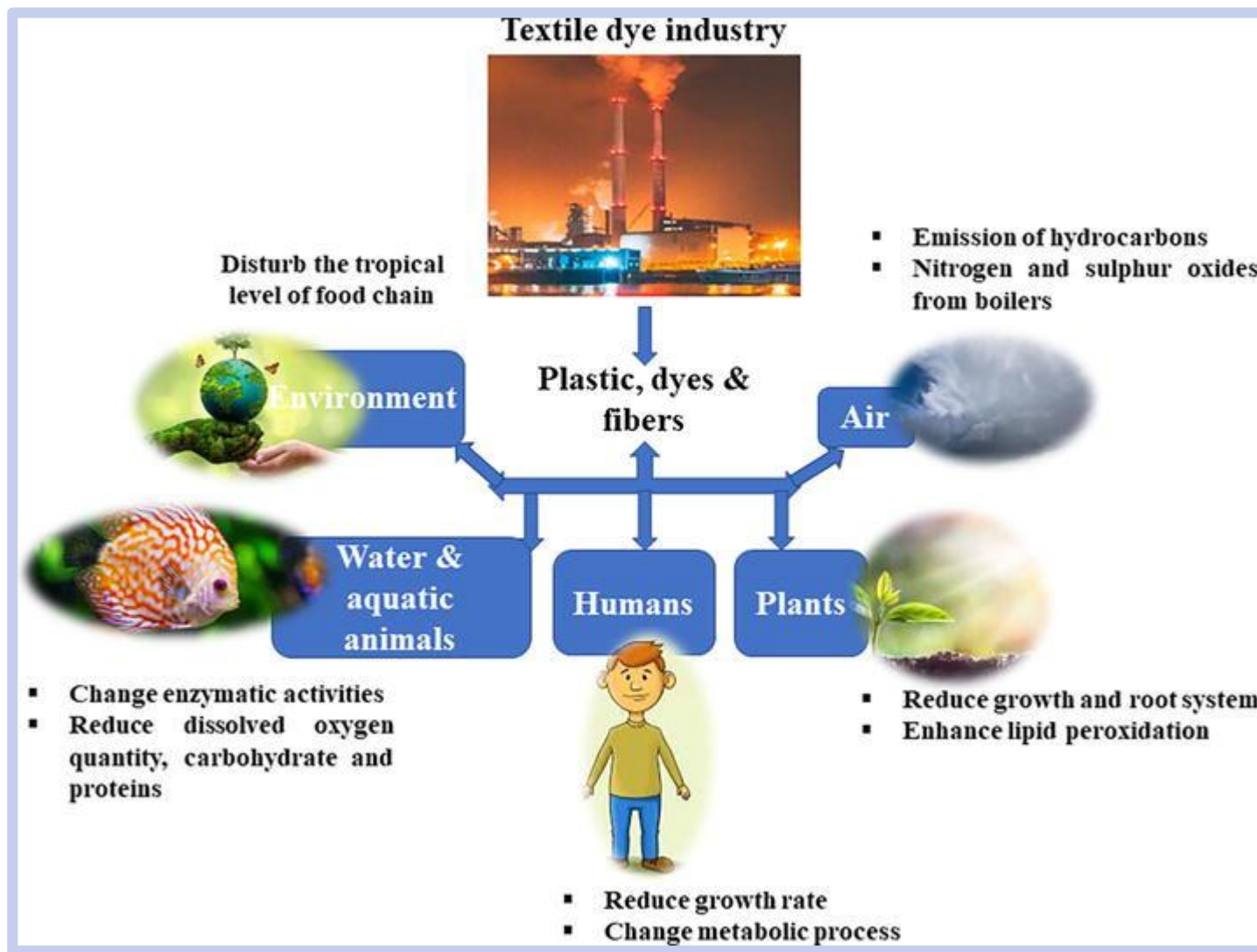


Figure 1: Impact of textile dyes on ecosystem (Subburaj et. al 2024)

To remove these compounds, a technique known as **biosorption** has been put into place. This technique involves utilizing biomass of organic substances, such as fungi or algae, in polluted areas to act as a sorbent.

The main disadvantage of biosorption as a technique is that spent biosorbent does not degrade sorbed dye particles. This poses an **environmental risk** as these sorbents still include the same toxins and carcinogens due to the dye particles' presence. It is vital to promote **the sustainable reuse and recycling** of spent adsorbents.

Zinc oxide nanoparticles (ZnO NPs) have photo-degradative properties that show promise for the sustainable elimination of toxic dyes. Incorporating the photodegradative properties of ZnO NPs into a *Pleurotus ostreatus* (oyster mushroom) biosorbent **can create a sorbent that can both sustainably degrade dye and be recycled for future biosorption**.

Purpose

At an industrial level, harmful chemicals (such as hydrochloric acid) and energy intensive processes are often employed to rid biosorbents of toxic dyes. However, zinc oxide nanoparticles (ZnO NPs) can be sustainably synthesized to degrade dye via photodegradation. Hence, **ZnO treatment could serve as a widespread biosorbent recycling method**. The combination of ZnO NPs and oyster mushroom is particularly useful because oyster mushroom is easily cultivated and commercially available.

Synthesizing an Enhanced *P. ostreatus* Biosorbent Utilizing Zinc Oxide Nanoparticles for Sustainable Wastewater Dye Degradation

Methods

Control Testing

Concentrations of methyl orange (MO) dye were tested



Figure 3: Spectrophotometric analysis of dyes

- 10, 20, 30, and 40 mg/L MO concentrations were tested for a spectrophotometric absorbance between 1 and 0
- 10mg/L MO solution was chosen for study

Oyster mushroom biosorbent was prepared



Figures 4 & 5: Crushing and drying of oyster mushroom

- Dried oyster mushroom was blended, washed and dehydrated
- Once the mushroom was dried, it was crushed again to form a fine powder

MO was adsorbed by dried oyster mushroom biosorbent



Figures 6 & 7: MO solution before and after 96h

- 2g of biosorbent was added to 200ml of 10mg/L MO solution
- Absorbance at 465nm after 96h was determined via spectrophotometry

Experimental Testing

ZnO-enhanced sorbent was prepared



Figures 8 & 9: Soaking and crushing of enhanced sorbent

- 2g of control sorbent was added to a 15% w/v ZnO NP dispersion in water
- Sorbent was soaked for 48h and dehydrated
- Final sorbent was crushed in a mortar and pestle to form a fine powder

Enhanced oyster mushroom sorbent was tested



Figure 10: Filtered MO solutions after 96hr

- 2g of control and enhanced sorbent were added to two 200mL 10 mg/L MO solutions
- Solutions were irradiated under sunlight for 96h
- Final absorbances were tested via spectrophotometry

Results

Control Sorbent Degradation (%)	Enhanced Sorbent Degradation (%)
81.3	5.74

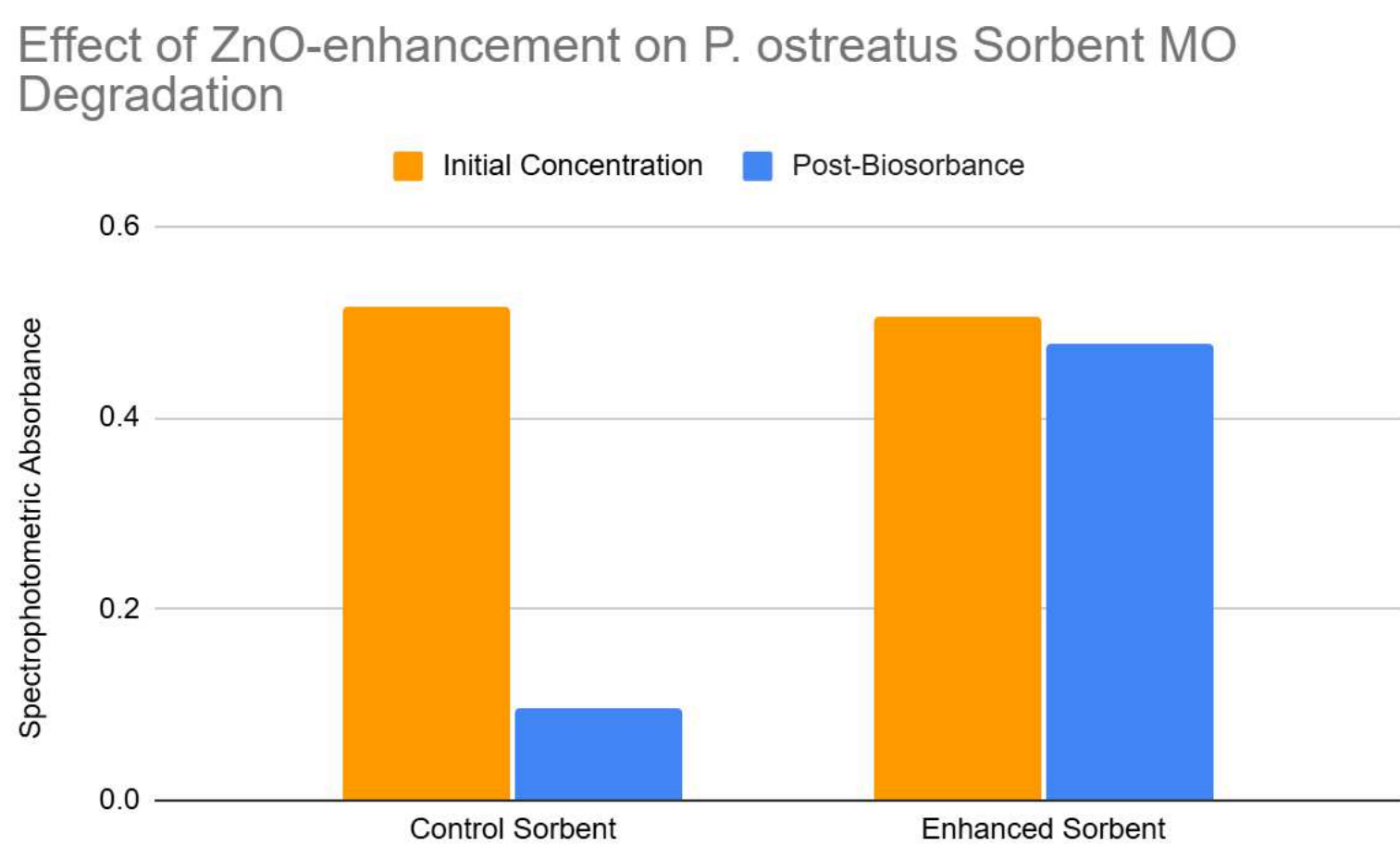


Figure 11: Absorbance graph

Conclusions

There is **no significant evidence** that the addition of ZnO NPs to the oyster mushroom biosorbent improved the MO degradation ability of the sorbent itself.

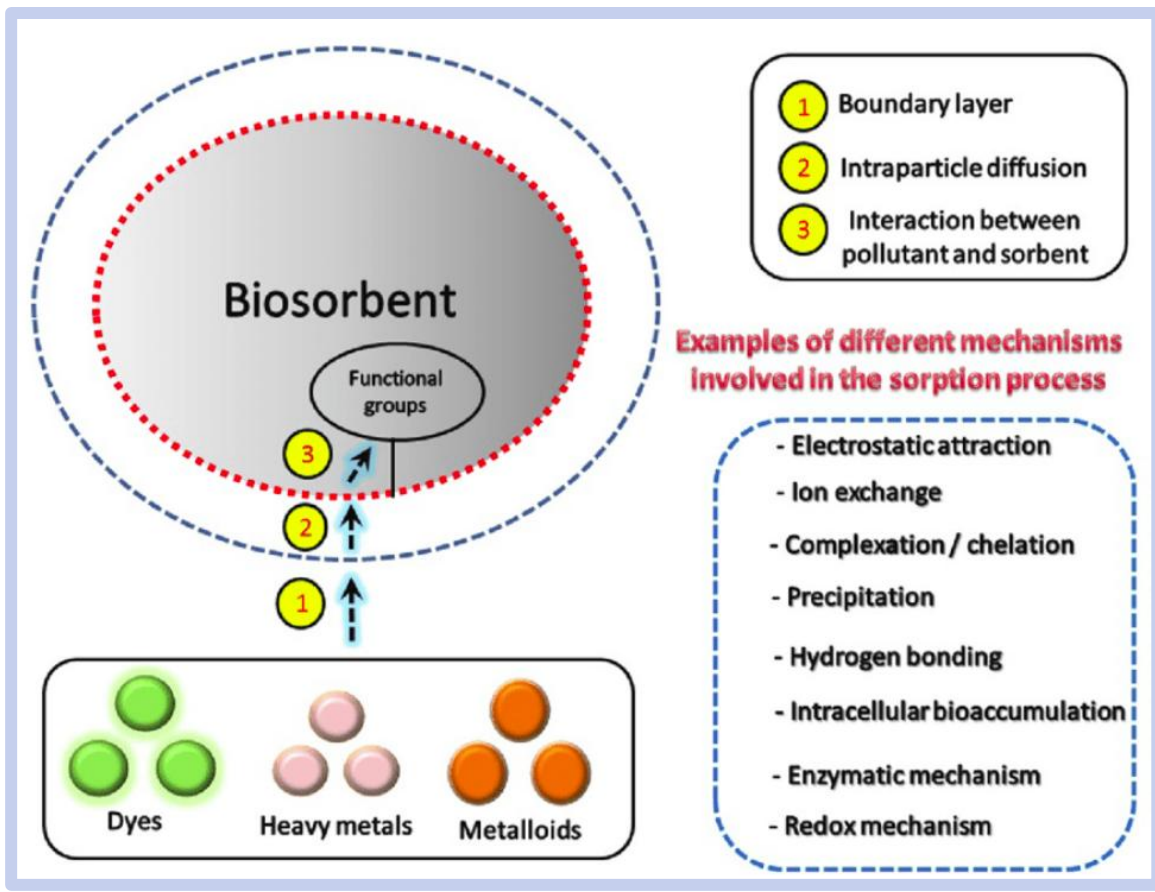


Figure 12: Mechanisms involved in the biosorption process. (Elgarahy et. al 2021)

Since ZnO is a polar material, it is hypothesized that its electrostatic attraction toward the surface of the oyster mushroom sorbent itself plugged the pores that the MO could have attached to.

As a result, the MO molecules may not have been able to bind themselves to the oyster mushroom because any potential binding sites were already occupied by the ZnO particles. This resulted in the **MO particles remaining in the enhanced sorbent mixture while the control MO particles were able to bind to the control sorbent**.

Applications

While ZnO NP modification may not be a viable way to enhance an oyster mushroom biosorbent, **other photo-degradative agents** could potentially be used to develop an enhanced biosorbent. For example, titanium dioxide and carbon nanotubes also have similar degradative properties to ZnO that could interact in different ways with the oyster mushroom sorbent to make it more effective.

In addition, **alternative methods of enhancing the sorbent could be researched**. Although a simple soak was utilized in the practiced methodology, advanced techniques such as crosslinking and polymerization of a hybrid degradation agent-sorbent material could be researched. Furthermore, because it was hypothesized that the ZnO NPs clogged the binding sites of the oyster mushroom biosorbent, different concentrations of ZnO NP dispersion could be tested to soak the sorbent in. It is possible that the dispersion used in this methodology was too concentrated and that a lower concentration would be optimal for degradation.

References

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